

LAB 13 REPORT

Name: Muhammad Sohaib

Reg No: 2026(S)CYS52

Lab Title: Recursion & Exception Handling

Theory: Recursion is a technique where a function calls itself to solve a problem by breaking it into smaller parts.

Exception Handling is used to handle errors in a program and prevent it from crashing using try and except blocks.

It helps create more reliable and error-free programs.

TASKS

TASK 1

```
def show(n):  
    print(n)  
show(5)
```

Output:

```
C:\Users  
5
```

TASK 2

```
def show(n):  
    if n==0:  
        return  
    print(n)  
    show(n-1)  
show(5)
```

Output:

```
5
4
3
2
1

Process finished with exit code 0
```

#TASK 3

```
def fact(n):
    if n==0 or n==1:
        return 1
    return n * fact(n - 1)
print(fact(6))
```

Output:

```
720

Process finished with exit code 0
```

```
# TASK 4
def fibonacci(n):
    if n==0:
        return 0
    elif n==1:
        return 1
    else:
        return fibonacci(n-1)+fibonacci(n-2)
for i in range(10):
    print(fibonacci(i))
```

Output:

```
0
1
1
2
3
5
8
13
21
34

Process finished with exit code 0
```

```
# TASK 5
```

```
try:  
    x = 10 / 0  
except ZeroDivisionError:  
    print("0 se divide nahi kar sakte")
```

Output:

```
0 se divide nahi kar sakte  
  
Process finished with exit code 0
```

```
# TASK 6
```

```
try:  
    num = int(input("Number: "))  
    data = [1, 2, 3]  
    print(data[num])  
except ValueError:  
    print("Number hi likhna tha")  
except IndexError:  
    print("List mein itne item nahi hain")
```

Output:

```
Number: 3  
List mein itne item nahi hain  
  
Process finished with exit code 0
```

